

Question

M is a gray-white metallic element, soluble in acids and bases, and highly toxic. It exists in nature in the form of silicate minerals with a +2 oxidation state. **M** reacts with chlorine gas to yield a white chloride **A**; **A** reacts with tetraphenylphosphonium chloride in a 1 : 2 molar ratio to yield **B**, while reacting in a 1 : 1 molar ratio yields **C**. **B** and **C** are both 2 : 1 type salts. When **A** is dissolved in a mixture of dinitrogen tetroxide and ethyl acetate at low temperature, **D** precipitates, containing only +1 valent cations and -2 valent anions, and no isolated solvent molecules; when **D** is heated to 50°C, a reddish-brown gas **E** is released, and **F** is obtained; when **F** is further heated, a mixture of gases **E** and **G** is released, and compound **H** is obtained; **H** is a tetranuclear complex with the same symmetry as a methane molecule.

- A.** All other options are incorrect.
- B.** Metal **M** is the metal with the weakest X-ray penetration ability.
- C.** The cations and anions in **B** possess different symmetries.
- D.** **B** and **C** have different coordination numbers for the metal **M**.
- E.** **D**, **F**, and **H** contain exactly the same elements.
- F.** **H** contains only one kind of bidentate ligand.
- G.** **E** and **G** are diamagnetic at room temperature.

Answer

Correct Answer: **E**

Detailed Explanation

1. Identification of Metal **M**

M is a gray-white metallic element, soluble in both acids and bases (amphoteric), and highly toxic. It exists in nature in the +2 oxidation state.

Among common amphoteric metals, Aluminum (Al) does not meet the oxidation state condition, Zinc (Zn) is not highly toxic, and Tin (Sn) exists in nature mostly in the +4 oxidation state, which does not meet the common natural oxidation state condition. Lead (Pb) meets the natural oxidation state condition, but it mainly exists as the sulfide PbS. None of these meet the requirements of the question.

At this point, consider the diagonal relationship. Due to the +2 valence state, consider Beryllium Be, the diagonal of Al, which is gray-white, amphoteric (soluble in acids and bases), highly toxic, and exists in nature in the +2 oxidation state (such as in beryl).

Therefore, **M** is beryllium (Be).

CHECKPOINT

1 PTS

Deduction: **M** is Be

2. Identification of Compound **A**

M(Be) reacts with chlorine (Cl₂): $\text{Be} + \text{Cl}_2 \rightarrow \text{BeCl}_2$.

A is a white chloride, consistent with the properties of BeCl₂ (white solid).

Therefore, **A** is BeCl_2 .

CHECKPOINT

1 PTS

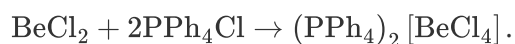
Obtained the chemical formula of **A**

3. Identification of Compounds **B** and **C**

A (BeCl_2) reacts with tetraphenylphosphonium chloride (PPh_4Cl).

PPh_4Cl provides PPh_4^+ and Cl^- , BeCl_2 is a Lewis acid and can form chloro complexes.

A 1 : 2 molar ratio reaction yields **B**:



The anion $[\text{BeCl}_4]^{2-}$ has a -2 charge, and the cation PPh_4^+ has a $+1$ charge, forming a 2 : 1 type salt (cation number:anion number = 2 : 1).

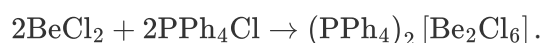
Therefore, **B** is $(\text{PPh}_4)_2[\text{BeCl}_4]$.

CHECKPOINT

1 PTS

Obtained the chemical formula of **B**

A 1 : 1 molar ratio reaction yields **C**:



BeCl_2 can dimerize in solution to form the $[\text{Be}_2\text{Cl}_6]^{2-}$ anion (containing a chlorine-bridged structure).

The anion $[\text{Be}_2\text{Cl}_6]^{2-}$ has a -2 charge, and the cation PPh_4^+ has a $+1$ charge, forming a $2 : 1$ type salt (cation number:anion number = $2 : 1$).

Therefore, **C** is $(\text{PPh}_4)_2[\text{Be}_2\text{Cl}_6]$.

CHECKPOINT

1 PTS

Obtained the chemical formula of **C**

4. Identification of Compound **D**

A (BeCl_2) is dissolved in a mixture of dinitrogen tetroxide (N_2O_4) and ethyl acetate, and **D** precipitates out at low temperature.

N_2O_4 dissociation: $\text{N}_2\text{O}_4 \rightleftharpoons \text{NO}^+ + \text{NO}_3^-$.

Ethyl acetate ($\text{CH}_3\text{COOC}_2\text{H}_5$) may participate in the reaction or hydrolysis to provide acetate (CH_3COO^-), but at low temperature, the main reaction is the

displacement of chloride by nitrate.

D contains only $+1$ valent cations and -2 valent anions, with no isolated solvent molecules.

The cation is NO^+ (from N_2O_4), and the anion is a -2 valent anion containing Be.

BeCl_2 reacts with NO_3^- to form $[\text{Be}(\text{NO}_3)_4]^{2-}$

Therefore, **D** is $(\text{NO})_2[\text{Be}(\text{NO}_3)_4]$, consistent with a $2 : 1$ type salt.

CHECKPOINT

1 PTS

Obtained the chemical formula of **D**

 5. Identification of Compounds **E**, **F**, **G**, and **H**

D is heated to 50°C to obtain **E** and **F**:

$(\text{NO})_2[\text{Be}(\text{NO}_3)_4]$ decomposes upon heating, releasing the reddish-brown gas NO_2 .

Therefore, **E** is NO_2 , and **F** is $\text{Be}(\text{NO}_3)_2$.

CHECKPOINT

1 PTS

Obtained the chemical formulas of **E** and **F**

Continued heating leads to the loss of NO_2 . Some of the nitrate ligands in the complex are converted to oxide ions. Furthermore, according to the question, this complex is similar to methane and has T_d symmetry. Therefore, the 4 Be atoms it contains are located at the vertices of a tetrahedron. From symmetry, the nitrate groups are distributed along the edges of the tetrahedron, with a total of 6. Valence calculation reveals that one O^{2-} ion can only be located at the center, coordinated to the 4 Be atoms. In this case, each Be atom is in a tetrahedral coordination environment, which is more reasonable.

Therefore, **H** is $\text{Be}_4\text{O}(\text{NO}_3)_6$.

CHECKPOINT

1 PTS

Obtained the chemical formula of **H**

The metal Be has the strongest X-ray transmission ability.

CHECKPOINT

1 PTS

B is incorrect.

The cation and anion in **B** are both tetrahedral and have the same symmetry.

CHECKPOINT

1 PTS

C is incorrect.

The metal Be in **B** and **C** are both tetracoordinate, with the same coordination number.

CHECKPOINT

1 PTS

D is incorrect.

D, **F**, and **H** contain exactly the same elements, namely Be, N, and O.

CHECKPOINT

1 PTS

E is correct.

H contains not only one type of bidentate ligand, but also an O^{2-} ion that can only be located at the center. The coordination number of this O^{2-} ion is 4.

CHECKPOINT

1 PTS

F is incorrect.

E and **G** both contain unpaired electrons, so they are both paramagnetic.

CHECKPOINT

1 PTS

G is incorrect.